Design and Implementation of a Scalable Field-Oriented Control Platform for BLDC Motors

Advanced Real-Time Systems - Final Presentation

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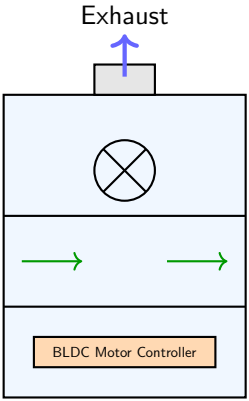
Case Study: Modern Ventilation Systems

The Industry Challenge:

- Growing demand for energy-efficient HVAC systems
- Need for precise airflow control
- Diverse applications: residential to industrial
- Requirement for flexible, scalable solutions

Our Approach:

- Design a **universal controller** platform
- Support multiple motor power ratings
- Implement advanced Field-Oriented Control (FOC)
- Validate across three motor classes

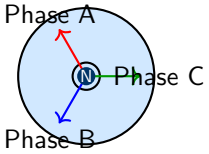


Smart Building

Why BLDC Motors in Ventilation Systems?

Advantages of BLDC:

- ✓ Higher efficiency (85-95% vs. 60-75%)
- ✓ Reduced maintenance (no brushes)
- ✓ Better controllability
- ✓ Longer lifetime
- ✓ Lower acoustic noise

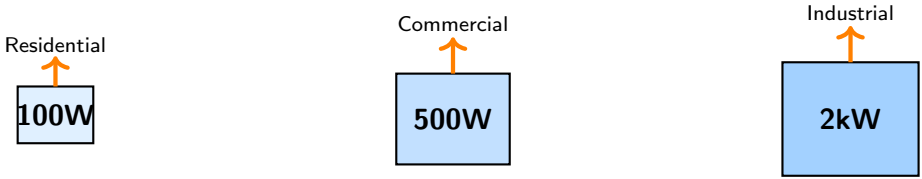


3-Phase BLDC Motor

The Challenge: Scalable Control Platform

Problem Statement

Ventilation system manufacturers need a **unified control architecture** that can drive motors from Low Wattage to High Wattage without separate controller implementations.



One Controller Architecture → Multiple Motor Classes

Project Objectives

1. **Comprehensive Plant Modeling**
   - Simscape Electrical BLDC motor model with parameterization
   - Three-phase electrical dynamics and mechanical coupling
2. **Field-Oriented Control Implementation**
   - Cascaded PI controllers for speed and current regulation
   - Clarke and Park coordinate transformations
3. **Scalability Demonstration**
   - Validated across Low Wattage to High Wattage motor range
   - Automatic controller gain scaling

Why Field-Oriented Control?

Traditional Control:

- × Control 3-phase AC currents directly
- × Coupled flux and torque
- × Complex non-linear dynamics
- × Difficult to tune

Field-Oriented Control:

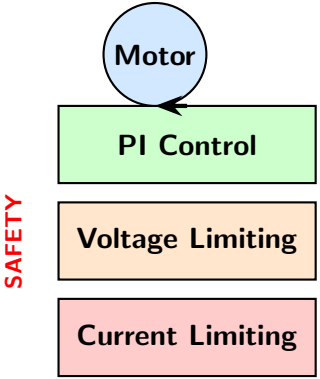
- ✓ Transform to rotating dq frame
- ✓ **Decouple** flux (I_d) and torque (I_q)
- ✓ AC $\rightarrow$ DC-like signals
- ✓ Simple PI control

Key Insight: AC currents become DC values in rotating frame – Easy PI control!

Controller Protection Features

1. **Current Limiting**
   - I_q saturated at motor-rated current
   - Prevents thermal damage
   - Critical for small motors

2. **Voltage Limiting**
   - V_d, V_q clamped to DC bus
   - Prevents over-modulation
   - Ensures safe operation



Protection ensures safe operation across multiple motor classes

Current Controllers (Inner Loop)

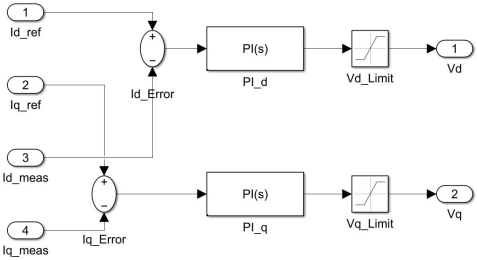


Figure: Parallel PI controllers for I_d and I_q regulation with voltage limiting.

Speed Controller (Outer Loop)

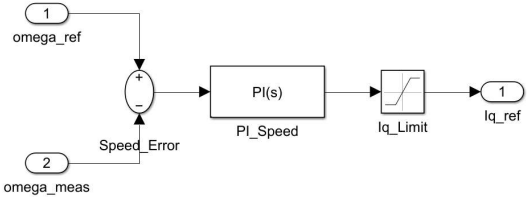


Figure: PI-based speed regulation with I_q reference generation and current limiting.

Complete FOC System Architecture

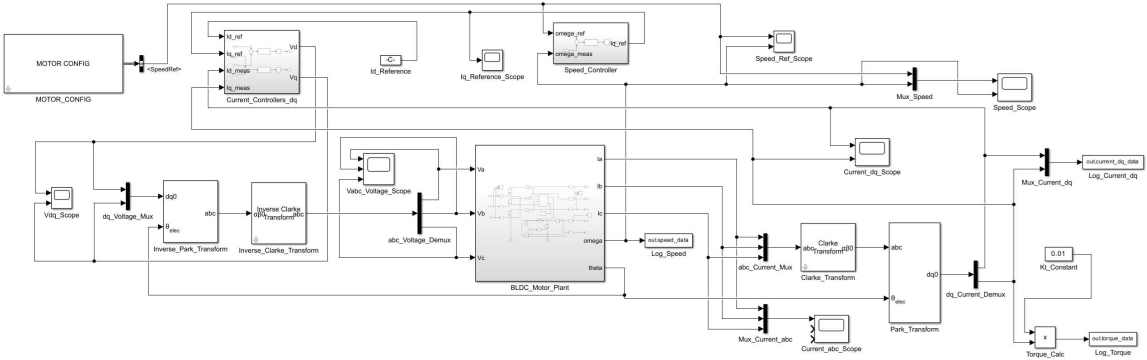
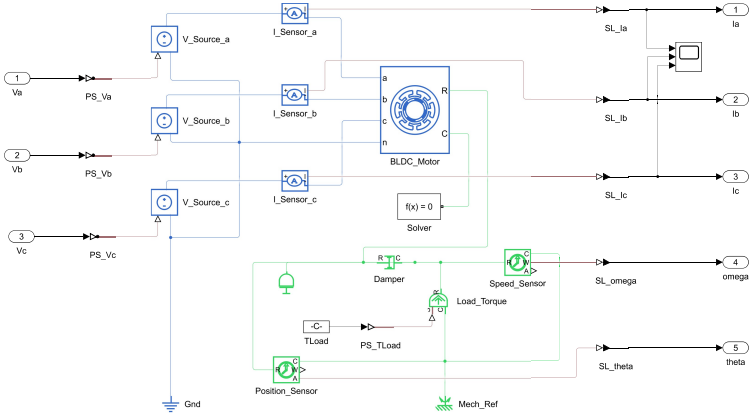


Figure: Complete Field-Oriented Control system showing cascaded control loops, coordinate transformations, and motor plant.

BLDC Motor Plant Model



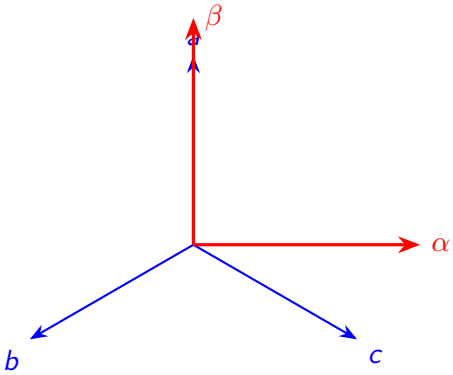
Clarke Transformation ($abc \rightarrow \alpha\beta$)

Goal: Dimensional Reduction

- Converts 3-phase quantities (abc) into a stationary 2-axis system ($\alpha\beta$)128.
- **Amplitude Invariant:** Preserves the magnitude of the vectors .

The Equation :

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$$



Stationary Frame

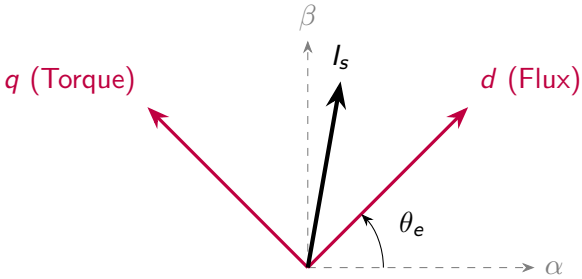
Park Transformation ($\alpha\beta \rightarrow dq$)

Goal: "Freezing" the Frame

- Rotates the $\alpha\beta$ frame to align with the rotor flux .
- Creates a **Rotating Reference Frame** ($d - q$).
- AC sine waves become constant **DC values**.

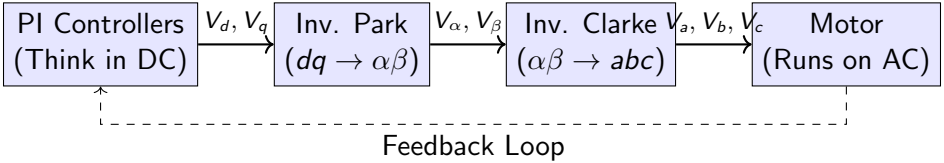
Decoupled Control:

- i_d (Direct): Controls **Flux**.
- i_q (Quadrature): Controls **Torque**.



Closing the Loop: Inverse Transformations

The Full Control Cycle



- **Inverse Park:** Un-rotates the voltage commands .
- **Inverse Clarke:** Converts back to 3-phase voltages for the inverter .
- **Computational Cost:** Transformations take $\approx 15\%$ of CPU time .

Simulation Configuration

Fixed-Step Solver Setup

- **Solver:** ode4 (Runge-Kutta 4th order) - excellent accuracy and stability
- **Timestep:** **10 μs fixed** - captures fast electrical dynamics

Why This Configuration Works Well:

- ✓ Provides deterministic timing behavior
- ✓ Captures electrical dynamics accurately
- ✓ Enables reliable validation before hardware deployment
- ✓ Ensures numerical stability for Simscape physical networks

This fine timestep ensures our system meets real-time performance requirements

Three Motor Classes

Parameter	**Small**	**Medium**	**Large**
Rated Voltage	24V	48V	96V
Rated Current	3A	10A	20A
Resistance (R_s)	2.5 Ω	0.8 Ω	0.2 Ω
Inductance (L_s)	5 mH	3 mH	1.5 mH
Torque Constant (K_t)	0.05	0.12	0.25
Inertia (J)	0.001	0.005	0.02 kg·m^2
Pole Pairs	4	6	8
Rated Speed	3000 RPM	2500 RPM	2000 RPM
Application	*Residential*	*Commercial*	*Industrial*

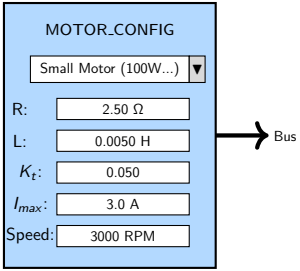
Table: Motor parameter specifications for three performance classes.

Key Insight: Same controller structure, automatically scaled gains!

Motor Configuration UI Subsystem

Interactive Parameter Selection:

1. **Dropdown Menu**
   - Three motor options with full specs
2. **Real-Time Display**
   - Shows R, L, K_t, I_{max}, rated speed
   - Updates instantly on selection
3. **Workspace Export**
   - Speed reference in rad/s
   - Electrical/mechanical conversion
4. **Bus-Based Distribution**
   - Single output bus
   - Bus Selector routes to blocks



Enables rapid motor switching for demonstrations without recompilation!

Automatic Controller Gain Scaling

How It Works:

- Controller gains are calculated automatically from motor parameters
- No manual retuning needed when switching motors
- Ensures optimal performance for each motor class

Current Controller:

- Gains based on motor inductance (L) and resistance (R)
- Smaller motors: Higher gains (faster response)
- Larger motors: Lower gains (stability)

Speed Controller:

- Gains based on motor inertia (J) and torque constant (Kt)
- Adapts to mechanical characteristics
- Maintains consistent damping across all motors

Current Regulation Performance

Three-Phase Currents:

- Balanced sinusoidal waveforms
- 120° phase separation

dq Components:

- $I_d \approx 0$ (proper FOC)
- I_q provides torque control

Key Technical Achievements

1. **Complete FOC Implementation**
   - Cascaded PI control with coordinate transformations
   - Decoupled flux and torque control
2. **Real-Time Validated**
   - ode4 solver with **mandatory 10 μs timestep**
3. **Robust Configuration Management**
   - Interactive motor configuration UI
   - Automatic gain scaling
4. **Safety and Protection**
   - Current limiting for thermal protection
   - Voltage limiting for safe operation

Project Benefits and Impact

For Manufacturers:

- Single validated controller
- Reduced development time
- Easier maintenance
- Scalable across product lines

For System Performance:

- Optimal efficiency
- Precise speed control
- Robust to disturbances
- Safe operation guaranteed

Foundation established for safe, efficient ventilation system motor control

Conclusions

Summary

This project successfully developed a **scalable Field-Oriented Control platform** for BLDC motors in ventilation systems, achieving:

- ✓ Complete FOC implementation with coordinate transformations
- ✓ Real-time validated execution (10 μs timestep)
- ✓ Scalable
- ✓ Interactive configuration
- ✓ Comprehensive safety and protection systems

Thank You!

Questions?

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